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TRACE-RAF: Trust-Gated Residual Analog Correction with Evidence-Preserving Retrieval for Auditable PM2.5 Forecasting

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ABSTRACT In the short horizon PM2.5 forecasting, the role of pollutant persistence, meteorology, calendar effects, as well as sparsity of environmental events and temporal shift in distribution, should be incorporated together without introducing leakage. In this study, a supervised TRACE-RAF, (Trust-gated Residual Analog Correction method) for the 24-hour forecasting with auditing is presented. The Source-Audited Context Builder integrates the EPA PM2.5 data, NOAA GHCNh weather, calendar and hash-checked NOAA Storm Events data. A Leakage-Safe Event-Context Retriever is leakage-safe because it enforces a candidate-target embargo; TRACE-RAF does not retrieve raw futures but, instead, out-of-fold base-model residual trajectories. Each correction is done by a reliability gate that scales it based on retrieval quality, model disagreement, correction spread, event context and distribution-shift evidence - all using 18 variables. A Drift-Aware Forecast and Evidence Head saves prediction level evidence and diagnostics. Evaluation includes Los Angeles County from 2019 to 2025 and 168-hour lookback and chronological splits of 45,784 origins of training, 7,950 validation, and 2,460 test. TRACE-RAF obtains MSE 40.062, MAE 4.134, and RMSE 6.329. It achieves 4.63% lower MSE than context XGBoost and is within 0.07% of context LightGBM, although Ridge achieves the lowest MSE of 39.144. Even though this nominal effect is not significant after Holm correction, TRACE-RAF reduces MAE by 0.021 compared to its base ensemble. Validation gives no weight to the raw-analogue fusion and to a global residual correction, while ungated residual transfer records MSE 44.959. The event-free TRACE-RAF variant is nearly identical, so an event-specific gain is not established. A significant degradation of the tested frozen Chronos-Bolt MAE is observed when conditioned on events, after multiplicity adjustment, which does not hold true for retrieval-augmented time-series foundation models. Since Storm Events doesn't have machine-readable publication times, all the results that are event-conditioned are still retrospective sensitivity analyses.

INDEX TERMS Air quality forecasting, PM2.5, retrieval-augmented forecasting, residual correction, time-series foundation models, distribution-shift diagnostics, auditable forecasting.

I. INTRODUCTION

THE fine particulate matter (PM2.5) is a significant urban air-quality risk factor, due to its association with respiratory and cardiovascular health impacts [1]. Short-term PM2.5 predictions can be useful for public-health alerts, reduction of exposure, planning of transportation, and environmental operations. However, in Los Angeles County, the dynamics of PM2.5 is not easily modeled using pollutant history. Higher and lower concentrations may occur due to wind speed, boundary-layer conditions, precipitation, holidays, wildfire smoke, and rare or unusual storm and high wind events during the season. All these make it a temporal and contextual problem in forecasting.

The recent time-series foundation models (TSFMs) hope to be able to transfer the learned forecasting capabilities to new domains by being pre-trained on a large time-series corpora to improve forecasting generalization. Examples are the models TimesFM, Moirai, MOMENT, Chronos and Tiny Time Mixers [2]–[6]. Meanwhile, non-parametric memory has also been demonstrated to provide useful external knowledge to foundation models during inference time (via retrieval-augmented generation) [7], [8]. TimeRAF brings these two together, and for zero-shot time-series forecasting, retrieves sequences from a curated knowledge base and then inject into a frozen TSFM via Channel Prompting [9].

The question of retrieval for air-quality forecasting involves a domain-specific retrieval question which is not fully answered by numerical time-series retrieval. A historical PM2.5 window can be one that is similar to the current window, but under different weather or event conditions. On the other hand, if a window is moderately numerically similar, but shares high wind, rain, holiday or stagnant or smoke context, the window is potentially more informative. This encourages a retrieval approach that makes records of sources as well as weather and calendar data and distribution-shift indicators part of the evidence for making forecasts.

This paper proposes a Trust-gated Residual Analog Correction mechanism (TRACE-RAF) for PM2.5 forecasting with event context as an auxiliary evidence channel under temporal distribution shift for auditable applications. The framework is based on retrieval augmentation in an environmental forecasting context and it does not share the same architecture as the learned retriever and Channel Prompting of TimeRAF. The implemented pipeline consists of four modules: Source-Audited Context Builder (SACB), Leakage-Safe Event-Context Retriever (LSER), TRACE residual-correction core and Drift-Aware Forecast and Evidence Head (DFEH). They collectively answer four questions: how official records can be reorganized to be retrieval compatible; how candidates can be selected without temporal leakage; how analogue residuals can correct a good supervised forecast without supplanting it; and, how each forecast can maintain evidence of retrieval, event, drift and uncertainty.

The main contributions are as follows:

- Source-Audited Context Builder: A reproducible mechanism to align EPA PM2.5, NOAA weather, calendar, and hash-verified NOAA Storm Events records, retaining source identity, coverage checks, and the declared event-availability assumption.
- Leakage-Safe Event-Context Retriever: Under the embargo where every candidate target ends before the query lookback begins, PM2.5 shape, weather, calendar, and event similarity are combined for historical analogue retrieval.
- TRACE residual correction: Out-of-fold residual memory and validation-trained reliability gate for selectively applying analogue corrections to a convex LightGBM–XGBoost context ensemble.
- Drift-Aware Forecast and Evidence Head: An evaluation layer that provides direct, residual-corrected, and frozen Chronos-Bolt paths, transparent shift indicators, probabilistic diagnostics, and machine-readable forecast evidence.

The evaluation is a final chronological holdout, and features linear, boosted tree, neural, retrieval-only, placebo-fusion and frozen-TSFM baselines. Paired inference is performed with 168-hour blocks, 5,000 bootstrap replications, HAC-corrected Diebold–Mariano tests and Holm adjustment for the set of comparison tests in the declared comparison family. This design allows you to see negative evidence as well as positive evidence and doesn't assume event association equals causation.

II. LITERATURE REVIEW AND RELATED WORK

A. STATISTICAL, MACHINE-LEARNING, AND DEEP FORECASTERS

Forecasting techniques vary in terms of inductive bias, data requirements, and computational requirements. Autoregressive and seasonal models are still useful baseline models due to straightforward failure modes. With an extension of this setting, tree ensembles like XGBoost and LightGBM can be used to model nonlinear lag, rolling, weather, and calendar interactions without the need for a sequence encoder [10], [11]. Recurrent LSTM and GRU models, on the other hand, acquire stateful temporal representations [12], [13]. However, systematic studies warn that one shouldn't blindly replace powerful statistical baselines with the recurrent networks, and they should be carefully preprocessed and evaluated [14]. DeepAR is an extension of the recurrent formulation to the probabilistic forecast of related series [15]. A set of large benchmark archives, such as the Monash Forecasting Repository, also allows for reproducible comparison of different heterogeneous domains of time-series [16]. Transformer families promote longer dependencies, including Informer with reduced attention cost, Autoformer and FEDformer with decomposition, Crossformer and iTransformer with cross-variable structure, and TimesNet with temporal variation reorganized to pattern in 2D [17]–[22]. The benefits of these gains do not mean that more complicated architecture is always required. Simple decomposition or patch-based inductive biases are shown to be still very competitive by DLinear [23] or PatchTST [24], while explicit multiscale decomposition and mixing are used by TimeMixer [25]. So, a deep backbone doesn't automatically win the evaluation; rather, the evaluation keeps the seasonal and boosted tree baselines. This option is in line with recent surveys of journals that focus on the fact that criterion for selecting a model should not be based on the novelty of the model structure but should consider the horizon, the structure of covariables, the uncertainty requirement, and the size of the data set [26], [27].

B. TIME-SERIES FOUNDATION MODELS

Instead of having to learn the representation of a time series from scratch, TSFMs aim for transfer of learning between time series. TimesFM features a decoder-only forecasting design, Moirai learns a universal forecasting Transformer, MOMENT can be used for a variety of time-series tasks, Chronos represents values as tokens, and Tiny Time Mixers focuses on compact zero- and few-shot transfer [2]–[6]. Timer, Timer-XL, UniTime, and Time-LLM are about generative pretraining, long-context scaling, crossdomain unification, or language-model reprogramming [28]–[31]. Frozen pretrained models have also been assessed for general computation engines for time-series analysis [32]. All of these studies encourage the study of zero-shot evaluation, yet they also demonstrate that input representation and domain shift affect the transfer performance. This way, the present study makes only comparisons on the same forecast origins between frozen Chronos-Bolt and locally supervised models, and does not presume the superiority of the foundation models.

C. RETRIEVAL-AUGMENTED FORECASTING

Retrieval augmentation is a disentanglement of parametric knowledge from an external memory. The concept of retrieve-then-condition set the example in natural-language processing for RAG and Dense Passage Retrieval [7], [8]. Diffusion forecasting of time series and time series assisted generation workflows have been explored as means of retrieval in time series [33], [34]. TimeRAF offers the most relevant methodological reference as they build a time-series knowledge base, train an MLP dual-encoder retriever, select top-k candidates and add them to a frozen TSFM by using Channel Prompting [9]. It ablations retrieval quality, integration, candidate no., knowledge-base composition and backbone choice.

The current framework is still based on the external-memory principle, but it poses a different experimental question. It relies on transparency (no learned dual encoder), random similarity, cosine similarity, and event-context similarity. Instead of using Channel Prompting, it uses a historical model residuals path and a learned trust gate as its main TRACE-RAF path, while its second path, Chronos-Bolt, evaluates output space fusion. It's not a copy of TimeRAF, therefore. It is an audited environmental adaptation to investigate the effects of contextual analogues as additional information after strong local forecasts have been generated.

D. AIR-QUALITY FORECASTING AND OFFICIAL CONTEXT

The pollutants history, temperature, humidity, wind, precipitation and pressure are all commonly used to help forecast PM2.5. Recent reviews report that a change has occurred from single-station recurrent models to attention-based, hybrid, and spatiotemporal models [35], [36]. For instance AirFormer, which models spatiotemporal evolutions deterministically at national scale and stochastically for uncertainty estimation [37]. This multi-station approach tackles spatial dependence and the current study analyzes an aggregate of counties and the origin of contextual evidence. Observations of regulated PM2.5 are provided by EPA AirData and hourly meteorology data are provided by NOAA GHCNh, while NOAA Storm Events provides structured records of hazards [38]–[40]. Smoke evidence could be added to NOAA Hazard Mapping System products but were not reported during the run [41].

E. DISTRIBUTION SHIFT AND FORECAST EVIDENCE

Means, variances and input-output relationships may change with temporal nonstationarity. Instead, TRACE-RAF tackles this issue by using interpretable shift indicators to audit regimes, without claiming shift has been eliminated, and by estimating statistics at local temporal scales [42]. Explanations can also be more or less general. SHAP attributes predictions to model parameters [43] and information-theoretic saliency allows checking counterfactuals of probabilistic predictions [44]. The evidence layer that has been implemented is more restricted: signed XGBoost effects, retrieved cases, event identifiers, drift components, proxy for uncertainty. While these records help traceability, they can not be used to prove causal effects of weather or events.

III. CRITICAL GAPS AND LIMITATIONS OF PREVIOUS STUDIES

Four gaps in the event sensitive urban air-quality forecasting are left by previous work. First, temporal forecasters and numerical retrieval systems do not always carry over the assumptions of provenance, coverage and availability of local contextual records. A numerically equivalent PM2.5 window can be present in a different weather or event regime and a retrospective event label may be confused with information provided at the time of forecast. The SACB answers to this gap, bringing together the official records under explicit source, coverage and availability audits.

Second, the historical retrieval may lead to the look-ahead bias if the candidate input or candidate target overlaps the query context. The fact that it is similar does not mean that the same future cannot be in retrieved evidence. The LSER fills this void with a temporal embargo, a training-history restriction, and separately auditable PM2.5, weather, calendar and event similarity terms.

Third, raw analogue futures can be redundant or contradictory to information that is already obtained by a robust supervised model. To overcome this gap, TRACE-RAF stores only strictly out-of-fold residual trajectories, and it learns a trust gate at the origin level from a validation-only correction utility. So it is asking whether an analogue accounts for the unexplained error and not whether the raw future should supplant the base forecast.

Fourth, retrieval gains are often reported without testing whether they remain stable when the distribution changes, are not due to a simple placebos of forecast-averaging, or are attributable to stored evidence. The DFEH fills this gap by using direct evidence, residual-corrected evidence, and frozen evidence of Chronos-Bolt paths on common origins, evidence that is both machine-readable and made available by the DFEH. This component provides comparison and auditability, but not a causal explanation that makes the contextual association.

These mappings delimit the novelty of the study. TRACE-RAF is lacking in a learned dual-encoder retriever as well as an inbuilt Channel Prompting and the current experiment on a single county does not provide for geographic generalization.

IV. CORE CONTRIBUTIONS

The proposed framework provides four mechanisms that have a one-to-one relationship with the gaps listed above:

- Source-Audited Context Builder (SACB): builds an aligned 85 dimensional context from official PM2.5, meteorological, calendar and event records, preserving the metadata of the sources and availability of these records.

- Leakage-Safe Event-Context Retriever (LSER): retrieves training-history analogues after enforcing a candidate-target embargo, and displays the contribution by each similarity channel and retrieved trajectory evidence.
- Trust-gated Residual Analog Correction (TRACE): provides an embargoed memory of out-of-fold forecast residuals, scales the retrieved residuals to match the query scales, and applies a validation-trained reliability gate instead of simply replacing the forecast.
- Drift-Aware Forecast and Evidence Head (DFEH): combines direct horizon models, TRACE-RAF, frozen-TSFM controls, drift-state analysis, and evidence of forecast in one chronological evaluation contract.

The empirical contribution is intentionally qualified. TRACE-RAF improves on its context ensemble in nominal MAE inference and remains competitive with the strongest nonlinear baseline, but the event channel itself has no verified all-origin effect and no favorable component comparison survives family-wise Holm correction.

V. PROBLEM FORMULATION

We will use x_t to represent the measurement of PM2.5 at time t. The air-quality lookback window for forecast time t is defined by the Equation (1).

$$X_t = [x_{t-L+1}, x_{t-L+2}, \ldots, x_t], \tag{1}$$

where L is the lookback length. We can now represent the weather covariates, W_t, with time aligned with the lookback window, calendar features, C_t, and event context, E_t is available at the point when the forecast is made, under the recorded source assumption. In Equation (2), it is the target that is defined for the input in Equation (1).

$$Y_{t+1:t+H} = [x_{t+1}, x_{t+2}, \ldots, x_{t+H}], \tag{2}$$

where, H is a forecasting horizon. Thus, Equation (2) represents the 24-step vector at each of the valid origins.

The forecasting function is given by the following Equation (3):

$$\hat{Y}_{t+1:t+H} = F_\theta(X_t, W_t, C_t, E_t, R_t^{ts}, R_t^{ev}, D_t), \tag{3}$$

where R_t^{ts} is the knowledge retrieved time-series, R_t^{ev} is the knowledge retrieved events, and D_t is the distribution-shift indicators. $L = 168$ hours and $H = 24$ hours are implemented target setting. The complete information interface is given in Equation (3) and the modules implemented to build the arguments are described in the section below.

The split based on time is given in Equation (4).

$$\mathcal{T}_{train} < \mathcal{T}_{val} < \mathcal{T}_{test}, \tag{4}$$

where training origins span January 8, 2019–August 23, 2024, validation origins span August 25, 2024–August 23, 2025, and test origins span August 25, 2025–December 30, 2025.

TABLE 1. Main Symbols Used in the Proposed Formulation

Symbol	Meaning
L	Lookback window length
H	Forecast horizon
X_t	Historical PM2.5 sequence at time t
W_t	Weather covariates
C_t	Calendar/context features
E_t	Event context available at the forecast origin
R_t^{ts}	Retrieved time-series knowledge
R_t^{ev}	Retrieved event knowledge
D_t	Drift indicators
q_t^B	Non-event base context in $\mathbb{R}^{46}$
q_t	SACB context vector in $\mathbb{R}^{85}$
G_t	LSER top-k retrieved set
$\rho(G_t)$	Retrieval summary in $\mathbb{R}^{51}$
$\hat{Y}_t^B$	Convex context-ensemble forecast
Δ_t	Retrieved residual correction in $\mathbb{R}^H$
z_t	TRACE reliability vector in $\mathbb{R}^{18}$
g_t	Validation-calibrated TRACE gate in $[0, 1]$
$\mathcal{O}_t$	Machine-readable DFEH evidence record
$\hat{Y}_{t+1:t+H}$	Predicted future PM2.5 values

These values of explicit cutoffs result in 45,784, 7,950, and 2,460 windows, respectively. For feature fitting the ordering is also kept in (4) as well as for retrieval eligibility, model selection and final evaluation; the test partition is not used to preserve a nominal percentage.

The basic notation has been summarized in Table 1.

VI. METHODOLOGY

The TRACE-RAF is carried out in 4 coordinated modules. The Source-Audited Context Builder (SACB) transforms the different types of official records into a chronological context with hourly time and preserves provenance and availability information. Leakage-Safe Event-Context Retriever (LSER) is used to build an embargoed historical memory and retrieve analogues based on PM2.5 similarity, weather similarity, calendar similarity and event similarity. The TRACE core maps residual memory from out-of-fold errors in a supervised context ensemble to an analogue memory and trains to recognize when the residual memory is valid. The Drift-Aware Forecast and Evidence Head (DFEH) reveals the final forecast, comparison paths, shift indicators and a machine-readable evidence record. No image operator, internal TSFM Channel Prompting or test-time parameter fitting is implied, all modules work with one dimensional time series or tabular features.

A. FRAMEWORK OVERVIEW AND FORWARD PASS

All the dataflow is summarized in figure 1. The first set of records to enter the SACB are EPA PM2.5, NOAA GHCNh weather, calendar and NOAA Storm Events records that are audited, aligned and transformed into query contexts, and input/target windows. The LSER is applied before the eligible candidates are ranked, so that the historical embargo is applied. At the same time, expanding-window XGBoost and LightGBM models generate strictly-out-of-fold (SOOF) residuals for the training memory. TRACE-RAF is able to retrieve and align those residuals, and estimate their reliability and add a gated correction to the context ensemble. Forecast, retrieval cases, event identifiers, drift evidence and uncertainty fields are recorded with the DFEH and frozen Chronos-Bolt is a separately evaluated benchmark path.

The full forecast for every origin t is given by Equation (5).

$$\hat{Y}_{t+1:t+H} = F_{\theta,\psi}\big(\phi(X_t, W_t, C_t, E_t), G_t, D_t, \mathcal{R}\big), \tag{5}$$

where $\phi(\cdot)$ is the tabular feature construction, G_t is the retrieved historical evidence, D_t is the drift evidence and R is an out-of-fold residual memory. The main methodological limitation is that all of the candidate targets and all of the residuals in G_t and R precede the start of the query lookback. The below forward pass is the expanded form of Equation (5).

The implemented forward pass is extended to the case of a batch of B origins in tensor notation as shown in Equation (6).

$$\begin{aligned}
Q^B &= \mathrm{SACB}_B(X, W, C) \in \mathbb{R}^{B\times 46},\\
Q &= [Q^B \,\|\, E] \in \mathbb{R}^{B\times 85},\\
G^r &= \mathrm{LSER}(Q, \mathcal{R}) \in \mathbb{R}^{B\times 8\times 24},\\
(\hat{Y}^L, \hat{Y}^X) &= \mathcal{H}(Q^B, C^+) \in (\mathbb{R}^{B\times 24})^2,\\
\hat{Y}^B &= \eta\hat{Y}^L + (1-\eta)\hat{Y}^X \in \mathbb{R}^{B\times 24},\\
(\Delta, Z) &= \mathcal{A}(G^r, \hat{Y}^B, D) \in \mathbb{R}^{B\times 24}\times\mathbb{R}^{B\times 18},\\
g &= \lambda\,\mathrm{clip}(h_\psi(Z), 0, 1) \in [0,1]^{B\times 1},\\
U &= \hat{Y}^B + g\odot\Delta \in \mathbb{R}^{B\times 24},\\
(\hat{Y}, \mathcal{O}) &= \mathrm{DFEH}(U, D, G^r) \in \mathbb{R}^{B\times 24}\times\mathcal{E}.
\end{aligned} \tag{6}$$

where $\eta = 0.6521$ is the out-of-fold-selected LightGBM weight, Δ is the aligned residual correction, Z is the 18 target-free reliability features, and $\lambda = 0.25$ is the validation-selected correction strength. E is the schema for an evidence record O. H is the paired context heads and A is the residual-alignment stage. The 23 PM2.5 features, the 14 weather features and the nine calendar features are present in Q^B while the 39 event features are included in Q but not in either of the base heads. There are 9 known future-calendar terms added to Q^B at each horizon, for a total of 55 inputs to each base regressor. The feature level M09 and the frozen Chronos-Bolt arms are not included in TRACE-RAF but will be carried out as a controlled comparison as defined by Equation (6).

B. SOURCE-AUDITED CONTEXT BUILDER

The SACB does so because of a desire to maintain data provenance and temporality prior to model fitting. In figure 2 you see that it is source specific checks before feature construction. EPA monitor data is rolled up to an hourly county median, NOAA weather data is synced from the weather station of choice, the event data is hash checked against an event manifest, and calendar information is calculated from the local time. The resulting context is then windowed and broken up chronologically—no interpolations are done for the target values.

1) Dataset Construction

The hourly PM2.5 data are collected from US EPA AirData/AQS parameter 88101 within the boundaries of Los Angeles County [38]. For coverage and continuity data, the data audit will pick one monitoring site where one monitoring site meets the coverage criterion. Since no one monitor qualified for this criterion, the criterion is the hourly median for all of the official Los Angeles County AQS monitors. Timestamps are converted to `America/Los_Angeles`, handling daylight-saving time explicitly. Target(s) are never interpolated into short input gaps, only observations from previous time periods can be used. The nearest appropriate NOAA Global Historical Climatology Network hourly (GHCNh) station [39] provides the weather covariates. Features of Calendar include cyclic hour, weekday, and month terms, weekend status, US federal holidays, and California holidays.

The Equation (7) specifies the concatenation of the group of features at each of the origins.

$$\begin{aligned}
q_t^B &= [p_t \,\|\, w_t \,\|\, c_t] \in \mathbb{R}^{23+14+9} = \mathbb{R}^{46},\\
q_t &= [p_t \,\|\, w_t \,\|\, c_t \,\|\, e_t] \in \mathbb{R}^{23+14+9+39} = \mathbb{R}^{85}.
\end{aligned} \tag{7}$$

The vectors p_t, w_t, c_t and e_t denote the PM2.5 feature vector, weather feature vector, calendar feature vector and event feature vector respectively. There are fixed lags, rolling moments and extrema, differences, the current value and a fill flag in the PM2.5 group. There are 24-hour rolling means of the weather group, which includes the measured variables, a precipitation-missing flag, derived condition flags, and a stagnation proxy. The event group includes the counts for 1 hour and active, 24/72/168 hours, 72 hours by event category and the event-burst ratio. Therefore, Equation (7) corrects both feature ordering, the 46 dimensional base context for C01, M04, C05 and C06 and the 85 dimensional event-aware context for LSER and event-conditioned controls.

Normalization statistics are only fit on the training split. To avoid this problem of differing scaling, the same preprocessing convention is used for query and candidate windows. The reported run contains 56,194 valid windows with X in $\mathbb{R}^{56194\times 168}$ and Y in $\mathbb{R}^{56194\times 24}$. Of these, 45,784 are allocated for training, 7,950 for validation and 2,460 for the final test holdout.

C. LEAKAGE-SAFE EVENT-CONTEXT RETRIEVER

Temporal eligibility is separated from relevance in the LSER. The embargo is first illustrated in Figure 3: a candidate will only be considered if its full target is before the start of the query lookback. The qualified candidates are then compared over four similarity channels that are stored separately. Future set is chosen to match the scale of the query & summarized for the forecasting head. This ordering is necessary so that such a high similarity score will not override the temporal constraint.

1) Time-Series Knowledge Base

PM2.5 windows from the past are stored in the time-series knowledge base. A window identifier, start time, end time,

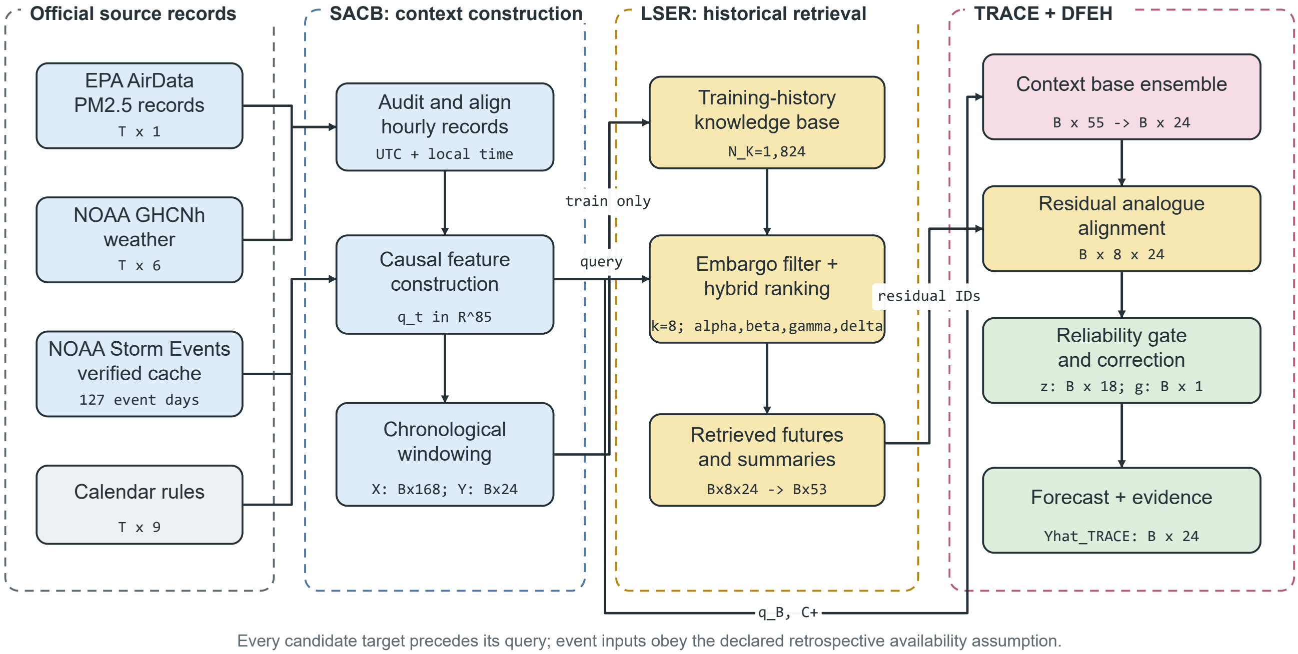

FIGURE 1. **Proposed TRACE-RAF pipeline. Official records pass through SACB, training-history candidates are filtered and ranked by LSER, out-of-fold residual analogues are scaled by the TRACE trust gate, and DFEH produces the 24-hour forecast with traceable evidence. Tensor annotations show the manifest-backed configuration.**

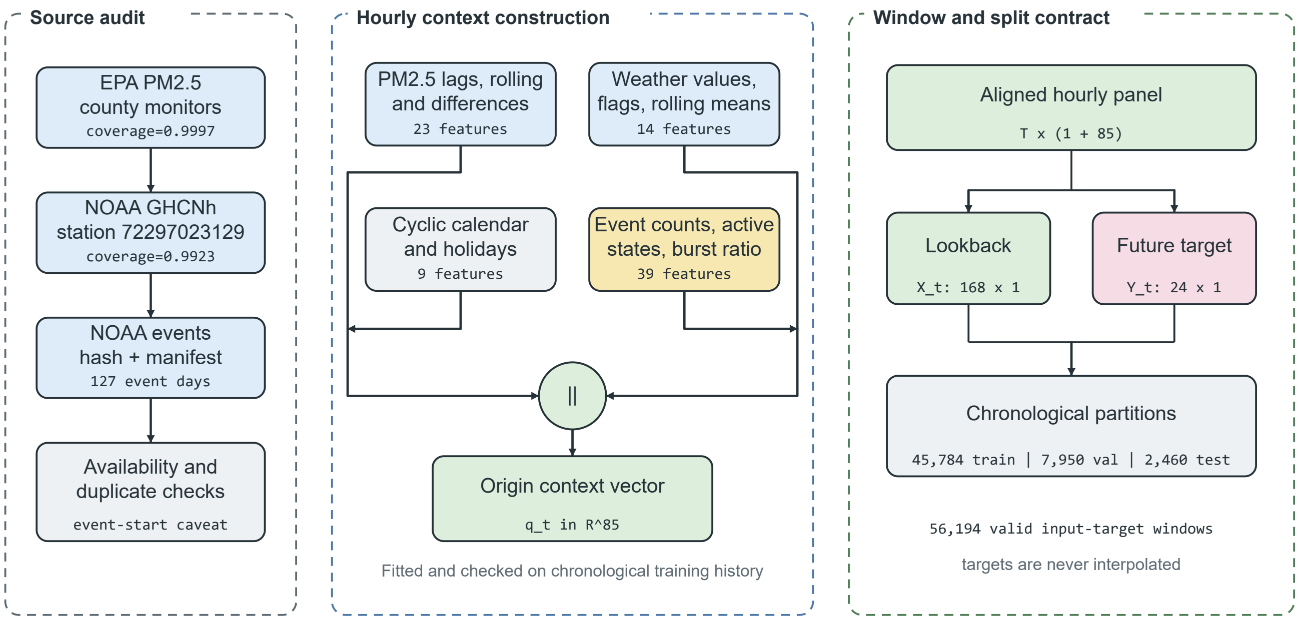

FIGURE 2. **Source-Audited Context Builder. Source checks feed four feature groups that form an 85-dimensional origin vector. The aligned panel is converted into 168-hour inputs and 24-hour targets before the chronological split.**

input values, future values, weather summary, calendar summary, normalized window vector, split and normalization identifier are stored in each record. The main knowledge base has 24 hours of origin stride, and 1824 training windows. The sensitivity arms take 6 hour and 1 hour strides and have 7,608 and 45,784 candidates respectively. The query specific

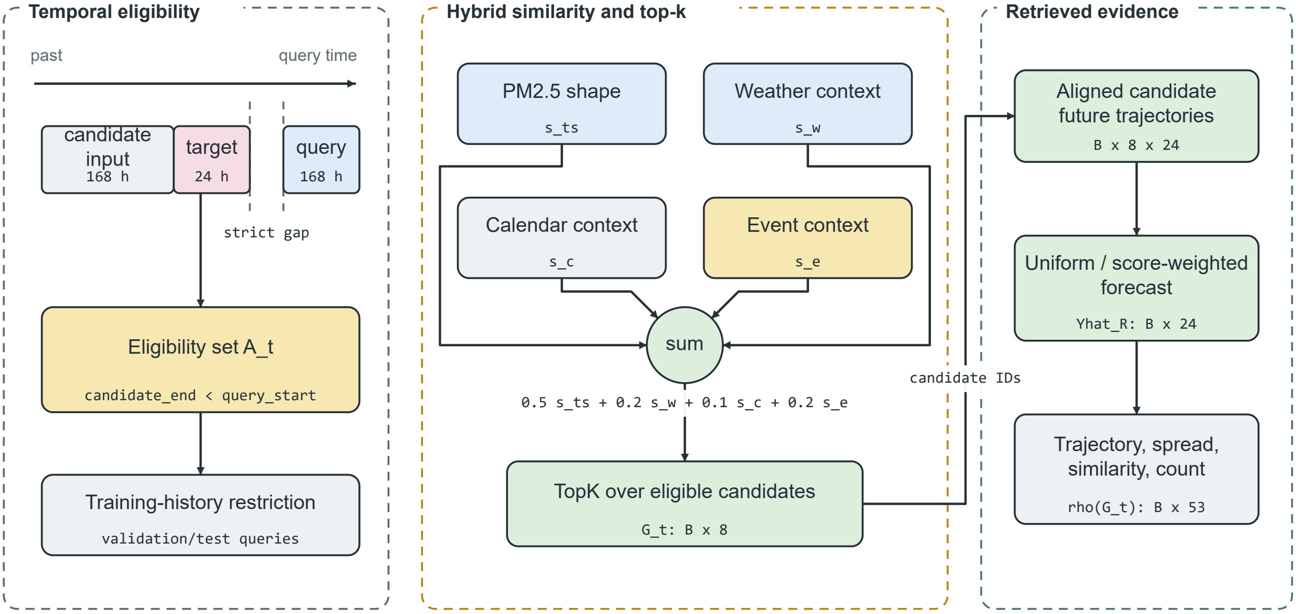

FIGURE 3. Leakage-Safe Event-Context Retriever. A strict candidate-target embargo and training-history restriction are applied before hybrid ranking. The top eight candidates provide aligned future trajectories and a 53-dimensional retrieval summary.

embargo in Equation (9) is not relaxed if there is overlap between the candidates.

Equation (8) formally defines the knowledge base.

$$\mathcal{K} = \{(u_i, X_i, Y_i, m_i, c_i, e_i)\}_{i=1}^{N}, \tag{8}$$

where u_i is the window candidate that defines the origin, X_i and Y_i are the window candidates that define the lookback and future windows, m_i is a weather summary, c_i is a calendar summary, and e_i is the window candidate that defines the event context within the origin window u_i. The set of equations which are temporally eligible for an origin t of a query are defined by Equation (9).

$$\mathcal{A}_t = \{i : u_i + H < t - L + 1,\ i \in \mathcal{T}_{train}\}, \tag{9}$$

Use for queries and test and validation. Strict inequality in (9) prevents that a candidate's target period ends after query input period starts. The training-history condition in (9) is also applied in the process of validation and test retrieval. Equation (8) and (9) represent the stored candidate schema and the admissible subset at query time, respectively.

The time-series retriever returns the records under query window X_t as defined in Equation (10).

$$R_t^{ts} = \{r_{t,1}^{ts}, r_{t,2}^{ts}, \ldots, r_{t,k}^{ts}\}, \tag{10}$$

where k is number of candidates retrieved. The default is k = 8, with sensitivity analysis for $k \in \{1, 4, 8, 16\}$. Equation (10) represents the returned records prior to summarization of the future trajectory of the records.

2) Event Knowledge Base

The event knowledge base consists of records about events noteworthy to Los Angeles County which are source preserving. While NOAA Hazard Mapping System fire/smoke records are still considered an unmeasured optional extension [40], [41], NOAA Storm Events is the required structured source. The information-availability assumption, location, category, source, source URL, title and summary are all stored fields. The reported records include information for the categories of wildfire, high wind, heavy rain, flood, and other weather categories – the slots for missing categories are set to zero.

The model inputs consist of hourly counts, 24-hour, 72-hour and 168-hour rolling counts, 72-hour counts by event category, counts of events in progress and an event-burst ratio. The more recent source identifiers are kept as forecast evidence. Target-horizon overlap is only a post hoc subset label and never is used as a model input. There is no machine-readable publication time for NOAA Storm Events, which is used in the implementation as a retrospective availability assumption. If the operation is very strict, it is necessary to have the records of the real publication or issue time.

3) Hybrid Ranking and Future Alignment

Both query and candidate PM2.5 windows are normalized by their respective means and standard deviations and unit length. Weather and calendar groups are standardized using candidate statistics, and their cosine similarities are mapped to $[0, 1]$. Event similarity is clipped to $[0, 1]$ with special handling of zero vectors. The hybrid retrieval score in Equation (11) includes all of the scores from the above: time-series similarity, weather relevance, calendar relevance and event relevance.

$$s(q,i) = \alpha s_{ts}(q,i) + \beta s_w(q,i) + \gamma s_c(q,i) + \delta s_e(q,i), \quad (11)$$

In which s_{ts} denotes time-series similarity, s_w denotes weather similarity, s_c denotes calendar similarity and s_e denotes event relevance. The implemented weights are $\alpha = 0.5$, $\beta = 0.2$, $\gamma = 0.1$, and $\delta = 0.2$. These are the primary weights which were predeclared and not a test-selected optimum, but Equation (11) still allows us to see each relevance channel so that we can consider the contribution of each without a learned latent scoring function.

The retrieval of the set is defined in Equation (12).

$$G_t = \mathrm{TopK}_{i\in\mathcal{A}_t}\ s(t,i). \quad (12)$$

A_t is only applied to Equation (12) after it's been evaluated. The restoration of future candidates to the query scale is defined in Equation (13).

$$\widetilde{Y}_{t,i} = \mu_t + \sigma_t\left(\frac{Y_i-\mu_i}{\max(\sigma_i,\epsilon)}\right), \qquad i\in G_t, \quad (13)$$

where μ_t, σ_t is query lookback statistics, and μ_i, σ_i is candidates lookback statistics. Forecasts are made on average, using retrieval only, uniformly or by non-negative normalized scores, $\widetilde{Y}_{t,i}$. The feature-level path is still the mean trajectory, horizon-wise spread, mean time-series similarity, maximum time-series similarity, and number of candidates, thus giving $\mathbb{R}^{51}$ for $\rho(G_t)$. The shape of the candidate trajectory is maintained by the affine transformation in Equation (13) which brings back the local scale of the query.

D. TRUST-GATED RESIDUAL ANALOG CORRECTION

TRACE-RAF alters the "what" that is retrieved. The part of the candidate future that is already predictable from the given context is dubbed the "raw" part. Therefore, adding such a future to the powerful supervised forecast can double count predictable structure. TRACE-RAF, on the other hand, builds a memory of out-of-fold residual trajectories and infers an origin-level trust coefficient for the retrieved correction. This remaining route of the two context models into the remaining memory, trust gate, final forecast and evidence record is depicted in figure 4.

1) Out-of-Fold Base Forecasts and Residual Memory

The base forecast is obtained by convex combination of the LightGBM and XGBoost context models as shown in Equation (14).

$$\hat{Y}_t^B = \eta\hat{Y}_t^L + (1-\eta)\hat{Y}_t^X, \qquad 0\le\eta\le 1, \quad (14)$$

where the fitted value of η is based on only the expanding-window out-of-fold (OOF) training predictions. There are two OOF prediction blocks: [0.50, 0.75) and [0.75, 1.00) of the training sequence. Each block has all the fitting targets finish prior to the start of the first prediction input. The audit shows 22,728 and 34,154 fit origins, 11,446 prediction origins/block, and a passed embargo in both cases.

The out-of-fold candidate's stored residual is defined by Equation (15).

$$\varepsilon_i = Y_i - \hat{Y}_i^{B,\mathrm{OOF}} \in \mathbb{R}^H. \quad (15)$$

The residuals are independently horizon clipped at the 0.01 and 0.99 quantiles of the training only data. There are 909 valid 24-hour residual-memory stride candidates, with additional 3,795 and 22,892 residual candidates at the 6-hour and 1-hour strides for evaluation during validation. The last stride is selected prior to test retrieval.

2) Residual Retrieval and Scale Alignment

LSER scores residual-memory candidates according to the same eligibility rule in time as Equation (9). Equation (16) combines the candidate residuals by transferring them into the query scale and uses the non-negative normalized retrieval weight $a_{t,i}$.

$$\Delta_{t,h} = \sum_{i\in G_t} a_{t,i}\frac{\sigma_t}{\max(\sigma_i,\epsilon)}\varepsilon_{i,h}, \qquad \sum_{i\in G_t} a_{t,i} = 1. \quad (16)$$

The same aligned residuals are used to give a spread vector per horizon. The difference between Equation (13) and Equation (16) is that the former passes a historical model error while the latter passes a historical target level.

3) Reliability Gate

For every origin, there is an 18-dimensional target-free vector z_t which contains the base-forecast mean, standard deviation and maximum; correction mean, standard deviation and maximum absolute magnitude; correction-spread mean and maximum; mean and maximum retrieval similarity; number of selected and eligible candidates using the log count; selected-event fraction and whether event conditioning was applied; mean and maximum LightGBM–XGBoost disagreement as well as the continuous and binary drift indicators. The ideal scalar correction coefficient which is used only during the validation is defined in Equation (17).

$$g_t^* = \mathrm{clip}\left(\frac{\langle Y_t - \hat{Y}_t^B, \Delta_t\rangle}{\|\Delta_t\|_2^2}, 0, 1\right). \quad (17)$$

This coefficient is learned by a depth-2 histogram gradient-boosting regressor $h_\psi(z_t)$. The first 67% of validation origins fit the provisional gate; the remaining 33% select λ in 0, 0.25, 0.5, 0.75, 1. The 24-hour residual-memory stride and $\lambda = 0.25$ are chosen, and the gate is then reattached to the entire validation partition (no test targets). The final forecast is given by Equation (18).

$$\hat{Y}_t^{\mathrm{TRACE}} = \hat{Y}_t^B + \lambda\,\mathrm{clip}(h_\psi(z_t),0,1)\Delta_t. \quad (18)$$

Equation (17) and (18) dis-entangle validation supervision from test time inference; that is, Y_t is needed during training of the gate but not used in any of the test reliability features.

E. DRIFT-AWARE FORECAST AND EVIDENCE HEAD

The DFEH is the common output and audit layer of the direct feature model, TRACE-RAF (validation only drift router), and the frozen Chronos-Bolt benchmark. The proposed path is an ensemble of contexts with a gated residual update as depicted in Figure 4. Separate branches maintain M09 feature-level conditioning and Chronos-Bolt forecast fusion to be able to test their effects independently of TRACE. Drift scores and flags remain as reliability inputs, subset definitions and evidence fields—the final TRACE gate is locked prior to opening the test partition.

1) Feature-Level Forecasting and Drift Diagnostics

The direct model input and prediction for the horizon h is defined by Equation (19).

$$u_{t,h} = [q_t \,\|\, \rho(G_t) \,\|\, d_t \,\|\, c_{t+h}], \qquad \hat{y}^X_{t+h} = f_h(u_{t,h}), \quad (19)$$

In $\mathbb{R}^{153}$ for the M09 comparison model and for each f_h is an XGBoost regressor trained with squared-error objective. This design permits using different supervised functions for each forecast horizon, but allows for a common origin representation. The 153 features verified so far (85 context, 53 retrieval, six drift and nine future-calendar values) are accounted for in Equation (19). The drift score and binary flag are only used to analyse the regimes and the evidence record and not to select the forecaster adaptively for the reported experiment.

2) Frozen-TSFM Forecast Fusion

Retrieval variant of the Chronos-Bolt that employs forecast-level fusion instead of internal Channel Prompting. Let $\hat{Y}^C_t$ be the frozen Chronos forecast and $\hat{Y}^R_t$ be the retrieved historical forecast. The fused prediction is given by Equation (20).

$$\hat{Y}^{CR}_t(\omega) = \omega \hat{Y}^C_t + (1-\omega)\hat{Y}^R_t, \quad (20)$$

where $\omega \in \{0, 0.25, 0.5, 0.75, 1.0\}$ is selected on the validation split. Validation selected $\omega = 0.25$. This branch does not modify Chronos-Bolt embeddings or weights since fusion takes place after both of the forecasts are made. So Equation (20) is not a prompting operation in the TSFM but an output space interpolation.

3) Distribution-Shift Diagnostics

The implementation is distribution shift and is done with transparent indicators and not claiming verified concept-drift identification. These indicators include the recent mean shift, recent variance shift, retrieval similarity drop, weather-pattern shift, and event-burst ratio. The raw drift vector which is used by the code is defined by Equation (21).

$$r_t = \left[|\mu_{24,t} - \mu_{168,t}|, \left| \log \frac{\sigma_{24,t} + \epsilon}{\sigma_{168,t} + \epsilon} \right|, 1 - \operatorname{clip}(\bar{s}_{ts,t}, 0, 1), \right.$$
$$\left. \sqrt{\frac{1}{4} \sum_{\ell=1}^{4} \left(\frac{w_{t,\ell} - \widetilde{w}^{tr}_{\ell}}{s^{tr}_{w,\ell}} \right)^2}, \ \max(b_{event,t}, 0) \right], \quad (21)$$

The standardised weather in the weather term are temperature, relative humidity, pressure and wind speed with respect to training medians and standard deviations. In Equation (22), each component $r_{t,j}$ is scaled using the training median and median absolute deviation (MAD), which is done robustly.

$$d_{t,j} = \frac{1}{6} \operatorname{clip}\left(\left| \frac{r_{t,j} - \operatorname{med}_{tr}(r_j)}{\max(1.4826\,\mathrm{MAD}_{tr}(r_j), \epsilon)} \right|, 0, 6 \right). \quad (22)$$

The continuous score and binary flag are defined in Equation (23).

$$S_t = \frac{1}{5} \sum_{j=1}^{5} d_{t,j}, \qquad D_t = \mathbb{I}[S_t \geq Q_{0.90}(\{S_v : v \in \mathcal{T}_{val}\})]. \quad (23)$$

Equation (21)–(23) are in line with the detector used: training data are used as the reference statistics, 0.90-quantile threshold is set using the validation data set, and the test data are transformed without refitting. The diagnostic regime analysis and forecast evidence are saved as a binary D_t.

4) Evidence Record and Uncertainty Proxy

The evidence module takes as input the signed contributions from local XGBoost models for each of the 24 direct models, weather flags, top three retrieved windows, calendar context, forecast direction, and component-level drift evidence. Its uncertainty proxy is given by Equation (24).

$$u^{\text{proxy}}_t = \sqrt{\bar{\sigma}^2_{R,t} + \bar{e}^2_{val}}, \quad (24)$$

The mean horizon-wise spread of the retrieved trajectories, $\bar{\sigma}_{R,t}$, and the validation residual RMSE, $\bar{e}_{val}$, are displayed. This means they produce a diagnostic scale rather than a calibrated prediction interval. Every evidence record is created from a stored field and not an open language model, and each of the event statements is still contextual and not causal. Only as an evidence-field definition, Equation (24) will not be used for the construction of confidence intervals.

F. RELATIONSHIP TO TIMERAF

TimeRAF leverages a learned MLP dual encoder to retrieve top-k time-series windows, and combines candidate embeddings with a frozen TSFM via Channel Prompting [9]. Table 2 lists the principles which are carried over and mechanisms which differ. This comparison does not rank the studies, but is methodological because of different datasets, horizons and protocols.

VII. EXPERIMENTAL SETUP

This section describes the final experiment named 20260827T04345754 which has a manifest. All the model, retrieval-density, fusion, gate and threshold selections were made before the final test partition was tested and were based on the training and validation data. Here the source, configuration, predictions, tables, evidence records, and figures have been archived as a package.

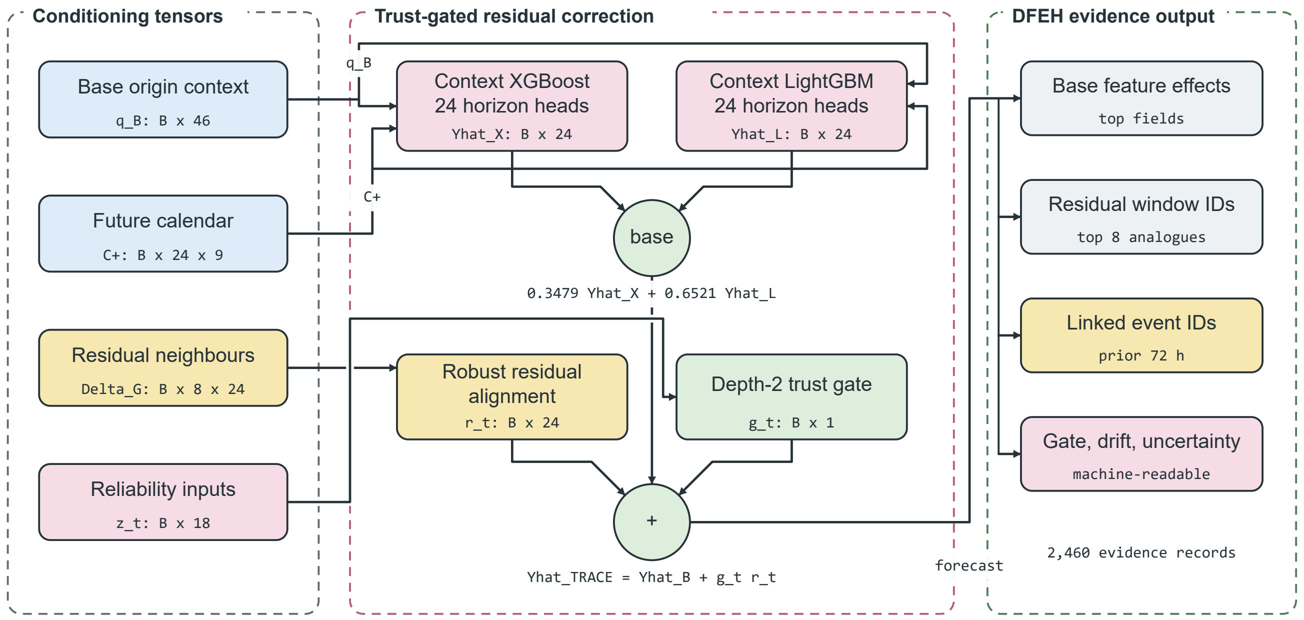

FIGURE 4. **TRACE-RAF and the Drift-Aware Forecast and Evidence Head. Out-of-fold LightGBM and XGBoost predictions form a context ensemble, LSER returns aligned residual analogues, and an 18-variable reliability gate scales the correction. The final forecast is accompanied by feature effects, retrieved cases, event identifiers, drift components, and uncertainty evidence.**

TABLE 2. **Methodological Comparison of TimeRAF and TRACE-RAF**

Aspect	TimeRAF	TRACE-RAF
Primary objective	Improve zero-shot forecasting by augmenting a frozen TSFM with external time-series knowledge.	Evaluate source-audited, event-context residual retrieval for local PM2.5 forecasting under temporal shift.
Forecasting domain	Multi-domain benchmark datasets.	Los Angeles County hourly PM2.5, 2019–2025.
Knowledge base	Curated, consistently normalized time-series windows.	1,824 primary training windows, dense-memory sweeps up to 45,784 windows, and 909 selected OOF residual candidates.
Retriever	Learned MLP dual encoder with forecaster-guided training.	Random, cosine, and transparent hybrid similarity; no learned retriever.
Knowledge integration	Internal Channel Prompting of candidate embeddings.	Trust-gated residual correction; feature- and forecast-level fusion are comparison arms.
Backbone	Frozen TSFM, principally TTM-Base.	Convex LightGBM–XGBoost context ensemble; frozen `amazon/chronos-bolt-small` is a benchmark.
Leakage control	Non-overlapping knowledge windows and cross-dataset training constraints.	Candidate target must end before query lookback; validation and test retrieval use training history only.
Event handling	Not an explicit module.	Availability-tagged NOAA Storm Events features and source-record evidence, reported retrospectively.
Drift and evidence	Retrieved-case inspection.	Five-component drift audit, reliability-gated residuals, signed feature effects, retrieved IDs, event IDs, and uncertainty evidence.
Implementation scope	Learned retriever and Channel Prompting are implemented.	Those mechanisms are not implemented and remain future extensions.

A. DATASET DESCRIPTION AND AUDIT

The study period is 2019–2025. The PM2.5 data were obtained from the US EPA AirData/AQS parameter 88101 for Los Angeles County [38]. The official Los Angeles County AQS target series is the hourly County median from official Los Angeles County AQS monitoring sites, not a single individual site. This fall-back selected `06-037-COUNTY` with 61,353 target hours observed, 191,284 contributing source records observed and observed coverage of 0.9998. The PM2.5 unit was always measured in $\mu g/m^3$.

Weather covariates were taken from NOAA GHCNh annual archives [39]. Station, Long Beach / Daugherty Field / Airport with station ID 72297023129, was selected which is located 10.36 km away from the centroid of the PM2.5 target area. The full weather-feature coverage post short-gap filling (allowing past feature) was 0.9923, while the raw measured core coverage was 0.9882. Every individual year's coverage was over 0.980 and the record was kept till 31 December

TABLE 3. Data-Readiness Audit

Gate	Value	Status
PM2.5 observed coverage	0.9998	Pass
Weather complete coverage	0.9923	Pass
Event days	127	Pass
Event overlap days	2,557	Pass
Qualifying event categories	4	Pass
Strict event availability	False	Caveat

2025.

Environmental events were downloaded from the NOAA Storm Events annual detail files verified using hash comparison [40]. This experiment used a locally staged cache of unaltered NOAA annual archives, and a source metadata file (a manifest) with official NOAA URLs, byte count, and official SHA-256 hashes since direct access to the NCEI in the execution environment was not available. The delivery mode used for the audit was `verified_official_noaa_cache`. The event audit identified 127 event days, 2,557 days of source overlap and four qualifying event categories. Records in the table are divided into categories of high-wind, flood, heavy-rain, other-weather and wildfire. Event-aware outputs assume that there is a retrospective availability of the event start time, which is not exposed by NOAA Storm Events. They don't necessarily predict an event in real-time. The resulting readiness status is as shown in Table 3. The source audit and the window contract are represented in Figure 2.

Table 4 summarizes the scope that has been evaluated and the contribution of each source in the forecasting pipeline.

The hourly source panel is used to reconcile with the model population in the origin-level attrition audit in Table 5. The most significant decrease is when it is required that all causal origin features have finite values; 56,194 of 61,177 bounded causal origin features are retained (91.85%). The 24 target values for a retained origin are directly observed.

B. DATA AND CODE AVAILABILITY

The raw public records and download pages are available from EPA AirData, NOAA GHCNh, and NOAA Storm Events [38]–[40]. NOAA Storm Events was not directly accessible in the experiment execution environment, so a locally staged NOAA Storm Events cache was used instead (all of the files in the NOAA Storm Events cache were verified against the created source manifest). You can find source code from https://github.com/shasabbir/event-time-raf. The implementation is done in Python 3.12.13 using the packages numpy, pandas, scikit-learn, xgboost, pytorch and chronos-forecasting; the pipeline and the configuration, notebooks, source modules and tests to reproduce the pipeline are provided if source files are publicly available.

C. FORECASTING TASK AND SPLIT

The forecasting task is designed to forecast PM2.5 one to 24 hours ahead and have a look-back window of 168 hours. The training set encompasses 45,784 origins from January 8, 2019 to August 23, 2024; validation has 7,950 origins from August 25, 2024 to August 23, 2025; and the frozen test holdout has 2,460 origins from August 25 to December 30, 2025. It will run through December 31, 2025 and generate 59,040 assessed hourly points for its targets. Random splitting is not done; test targets are not used to select the model or threshold.

D. MODELS

The base set includes seasonal rules, regularized linear regression, boosted trees, neural sequence models, retrieval-only forecasts, and a "frozen" TSFM. A comparison between the proposed M13 TRACE-RAF model, its C06 context ensemble and the event-free A03 gate. The drift in null hypothesis p-values presented in this report is not a post-hoc effect but rather a drift router separately selected by M12. Chronos placebos combine the same frozen TSFM forecast with climatology and persistence, enabling the averaging of forecast information to be separated from the retrieval. The foundation-model evaluation is carried out by `amazon/chronos-bolt-small`. The assessed configurations are listed in Table 6 according to their forecasting/retrieval functions.

E. IMPLEMENTATION AND HYPERPARAMETERS

The experimental setup was implemented on cloud with python 3.12.13 and Linux 6.12.90. The publication pipeline spent a total of 7,432.7 seconds in the pipeline. The main packages that are documented in the run manifest are NumPy 2.0.2, pandas 2.3.3, scikit-learn 1.6.1, XGBoost 3.2.0, LightGBM 4.6.0, PyTorch 2.10.0+cu128, Transformers 4.57.6, and chronos-forecasting 2.3.1. The main hyperparameters are listed in table VII.

F. EVALUATION METRICS AND STATISTICAL CHECKS

The main metric measurements are MSE, MAE, RMSE, MAPE, sMAPE and R^2. MSE is still used as the main squared error metric, with MAE and RMSE offering intuitive error scales for PM2.5. The most common error measures are given in Equation (25) for n observations y_j of a forecast and $\hat{y}_j$ of predictions.

$$\begin{aligned} \mathrm{MSE} &= \frac{1}{n}\sum_{j=1}^{n}(y_j-\hat{y}_j)^2, \\ \mathrm{MAE} &= \frac{1}{n}\sum_{j=1}^{n}|y_j-\hat{y}_j|, \\ \mathrm{RMSE} &= \sqrt{\mathrm{MSE}}, \end{aligned} \tag{25}$$

Equation (26) summarizes explained variance.

$$R^2 = 1 - \frac{\sum_{j=1}^{n}(y_j-\hat{y}_j)^2}{\sum_{j=1}^{n}(y_j-\bar{y})^2}. \tag{26}$$

The overall result is obtained by aggregating the forecasts across all forecast origins and horizons as in Equation (25), while Equation (26) uses the same set of points. MAPE and sMAPE are provided in machine-readable outputs but are not

TABLE 4. Dataset and Source Summary

Data group	Source	Evaluated scope	Use in pipeline
PM2.5	EPA AirData/AQS hourly parameter 88101	Los Angeles County, 2019–2025; 61,353 observed target hours	Target sequence and lag/rolling features
Weather	NOAA GHCNh, station 72297023129	2019–2025; 0.9923 usable core coverage	Temperature, humidity, wind, pressure, and rain features
Events	NOAA Storm Events annual detail archives	127 event days; four qualifying categories	Retrospective event-start features and forecast evidence
Calendar	Generated from timestamps and public holiday rules	2019–2025 hourly calendar	Hour, weekday, weekend, month, season, and holiday features

TABLE 5. Forecast-Origin Attrition Audit

Stage	Origins	Retained (%)
Bounded candidate origins	61,177	100.00
Chronological split eligible	61,129	99.92
Complete 168-hour lookback	61,121	99.91
Complete observed target	60,953	99.63
Complete causal origin features	56,194	91.85
Final retained windows	56,194	91.85

used for the main ranking due to unstable percentage errors when PM2.5 values are low. Subset metrics are reported only when at least 50 forecast origins are available. The 72-hour event-context subset contains 70 origins and the drift subset contains 407; the 33 target-event and eight active-event origins are retained in counts but suppressed from performance tables. Pairwise comparisons use 5,000 paired moving-block bootstrap resamples with 168-hour blocks and Diebold–Mariano tests with 168 HAC lags. Both raw and Holm-adjusted p-values are stored. AQI threshold decisions are evaluated at 35.4 and 55.4 $\mu\text{g}/\text{m}^3$, while Chronos quantiles are evaluated by interval coverage, an approximate quantile-grid CRPS, and calibration error.

G. COMPARISON WITH EXISTING STUDIES

Because of the differences in datasets, forecast horizons, variables and evaluation protocols, direct numerical comparison of the results of this paper with existing TSFM and air-quality papers is not possible. Thus, Table 8 is a comparison of methodological coverage, not a dominance in performance across papers.

VIII. RESULTS AND DISCUSSION

A. OVERALL COMPARISON WITH BASELINES

The results for the final-holdout are presented in Table 9. The ridge regression with context features has the best overall performance with MSE of 39.144, MAE of 4.079, RMSE of 6.257, and $R^2 = 0.457$. TRACE-RAF obtains MSE 40.062, MAE 4.134, and RMSE 6.329. It is 4.63% less than MSE of M04 XGBoost context and is just 0.07% more than C05 LightGBM context, but it is less than Ridge and LSTM. This ranking is significant: The proposed correction is a competitor to nonlinear context models, and the smooth county-median target and late-2025 holdout are good for regularized linear structure.

The frozen Chronos-Bolt evaluation completed successfully and validation selected a TSFM weight of 0.25. Event-retrieval fusion increases MSE from 47.534 to 48.609 and MAE from 4.202 to 4.574. The MAE increase of 0.372 remains significant after Holm adjustment (adjusted bootstrap p = 0.046; adjusted DM p = 0.035). Furthermore, the persistence-fusion placebo reaches MSE 46.964, lower than both M10 and M11. In this way, this run provides evidence against attributing a generic averaging effect to event retrieval. Chronos native quantiles remain useful diagnostically: M10 attains approximate CRPS 3.187, mean absolute calibration error 0.017, and 0.761 empirical coverage for the nominal 0.80 interval, whereas M11 degrades those values to 4.120, 0.176, and 0.206.

B. ABLATION STUDY

Table 10 dissociates the residual-memory hypothesis from its integration rule. All the selected values are on the last 2,624 origins of the validation partition starting on May 3, 2025 and excluding the test partition. The raw-future fusion (A04) and the global residual scale (A06) both pick zero, and reduce exactly to C06. This is not a failure of implementation, it's a validation outcome. The fixed (ungated) residual update (A05) has a MSE of 44.959 whilst the learned origin-specific gate has strength of 0.25 and a MSE of 40.062. Therefore, ablation will confirm the gate's conservative control function but not a statement that there is always a benefit from a residual retrieval.

The paired MSE contrasts declared in Table 11 favor the first model in the pair (negative differences). The nominal M04 improvements are for the cosine retrieval features and the matched feature-count placebo was nominally outperformed by the raw event features. The MSE of TRACE-RAF is almost the same as its base ensemble and LightGBM. Its MAE differences are -0.021 versus C06 (unadjusted 95% block-bootstrap CI [-0.038, -0.005]) and -0.017 versus C05 (unadjusted 95% block-bootstrap CI [-0.031, -0.001]). The respective Holm adjusted bootstrap and DM p-values are 0.355 and 0.490, and 0.381 and 0.516. There is no difference between the M13-A03 contrast (which is near zero), demonstrating that event conditioning was not directly revealed by the aggregation of TRACE-RAF's result. After applying

TABLE 6. Evaluated Model Families

ID	Description
C00–C06	Hour–month climatology; Ridge and LightGBM context; calendar and event retrieval; raw-event XGBoost; convex context ensemble
M00–M02	Persistence, daily seasonal, and weekly seasonal naive baselines
M03–M09	PM2.5/context XGBoost; retrieval-only controls; feature-level retrieval, event, and drift variants
B00–B02	DLinear, PatchTST, and LSTM trained on the same PM2.5 windows
M10–M12	Frozen Chronos-Bolt, Chronos retrieval fusion, and validation-only drift router
M13	Proposed TRACE-RAF trust-gated residual analogue correction
A00–A03	Event-free, random-retrieval, matched feature-count placebo, and event-free TRACE ablations
P00–P01	Chronos fused with climatology or persistence placebos

TABLE 7. Experimental Hyperparameters and Runtime Settings

Component	Setting
Study period	2019–2025
Lookback and horizon	$L = 168$ hours, $H = 24$ hours
Split	45,784 train; 7,950 validation; 2,460 final-test origins
SACB feature groups	23 PM2.5, 14 weather, 9 calendar, 39 event features
SACB missing-data rule	Past-only filling for gaps up to 3 hours; no target interpolation
LSER retrieval	$k = 8$, sensitivity for $k \in \{1, 4, 8, 16\}$
LSER knowledge base	1,824 windows at 24-hour primary stride; 6/1-hour sweeps
LSER similarity weights	time series 0.5, weather 0.2, calendar 0.1, event 0.2 (predeclared primary)
DFEH direct input	153 features for each of 24 horizon regressors
DFEH XGBoost	350 estimators, max depth 6, learning rate 0.05, subsample 0.85, colsample 0.85, $\lambda = 1.0$
TRACE base ensemble	LightGBM weight 0.6521; XGBoost weight 0.3479
TRACE residual memory	OOF boundaries 0.50/0.75/1.00; 909 selected candidates
TRACE trust gate	18 inputs; depth 2; 100 iterations; learning rate 0.05; selected strength 0.25
DFEH drift detector	5 components; 24-hour recent window, 168-hour reference window, 0.90 validation quantile threshold
DFEH Chronos path	`amazon/chronos-bolt-small`, batch size 64
DFEH fusion weights	$\{0, 0.25, 0.5, 0.75, 1.0\}$; selected $\omega = 0.25$
Neural optimization	AdamW; seed 42; batch 256; at most 30 epochs; patience 5; learning rate 0.001; weight decay 0.0001
DLinear	moving-average kernel 25; selected epoch 6
LSTM	hidden size 64; 2 layers; dropout 0.1; selected epoch 2
PatchTST	patch/stride 24/12; 3 layers; $d = 64$; 4 heads; FFN 128; selected epoch 4
Forecast evidence record	Top 3 feature effects, windows, and recent event IDs
Inference	5,000 paired resamples; 168-hour blocks/HAC lags; Holm correction

TABLE 8. Methodological Comparison with Related Forecasting Work

Method family	TSFM	Retrieval	Events	Evidence	Comment
PatchTST-style Transformers [24]	No	No	No	Limited	Strong sequence modeling, but event context is not central.
Chronos [5]	Yes	No	No	Limited	Frozen zero-shot baseline used in this study.
TimeRAF [9]	Yes	Yes	No	Retrieval cases	Base retrieval-augmented TSFM method; does not target official event-context PM2.5.
XGBoost context baseline [10]	No	No	No	Feature importance	Strong supervised tabular baseline for local PM2.5.
TRACE-RAF	No[a]	Yes	Yes	Evidence-grounded	Adds official-source audit, OOF residual retrieval, a trust gate, drift diagnostics, and stored evidence.

[a]Chronos-Bolt is evaluated as a frozen benchmark, but the proposed TRACE-RAF forecast is a supervised residual-correction model rather than a foundation model.

Holm corrections for the entire 29-comparison family for each metric, only the negative M11–M10 MAE contrast is significant.

The ablation does not form a monotone accuracy progression. Validation faithfulness is mainly influenced by PM2.5 history, weather and calendar features. The paper thus concludes that the behavior of TRACE-RAF is not due to a proven general event effect, but due to selective residual correction.

The results of the k sensitivity test presented in Table 12 indicate that within the tested range, the more neighbors that are included for the retrieval-only cosine baseline, the more accurate the baseline is. The best retrieval-only configuration among $k \in \{1, 4, 8, 16\}$ is k = 16, with MSE 51.316. The predefined default value of k = 8 was used for the feature models reported.

TABLE 9. Manifest-Backed Overall Test Results for 24-Hour PM2.5 Forecasting

Model	MSE	MAE	RMSE	R^2
C00 hour–month climatology	73.915	6.398	8.597	-0.025
M00 persistence	57.327	4.840	7.571	0.205
M03 XGBoost PM2.5	44.547	4.317	6.674	0.382
M04 XGBoost context	42.006	4.233	6.481	0.417
C01 Ridge context	**39.144**	**4.079**	**6.257**	**0.457**
C05 LightGBM context	40.089	4.151	6.332	0.444
B00 DLinear	40.305	4.097	6.349	0.441
B01 PatchTST	40.766	4.111	6.385	0.435
B02 LSTM	39.978	4.095	6.323	0.445
C06 context ensemble	40.115	4.156	6.334	0.444
M09 event-feature XGBoost	41.193	4.148	6.418	0.429
M13 TRACE-RAF	*40.062*	*4.134*	*6.329*	*0.444*
A03 TRACE-RAF without events	40.062	4.135	6.329	0.444
M10 frozen Chronos-Bolt	47.534	4.202	6.894	0.341
M11 Chronos + event retrieval	48.609	4.574	6.972	0.326
M12 validation drift router	41.466	4.190	6.439	0.425

TABLE 10. Validation-Selected TRACE-RAF Mechanism Ladder on the Final Holdout

ID	Forecast integration	Retrieval target	Integration control	MSE	MAE
C06	Base ensemble	None	None	40.115	4.156
A04	Base + raw analogue	Raw future	Weight 0.00 (validation)	40.115	4.156
A05	Base + residual analogue	OOF residual	Strength 1.00 (fixed)	44.959	4.250
A06	Base + residual analogue	OOF residual	Strength 0.00 (validation)	40.115	4.156
M13	Base + gated residual analogue	OOF residual	Gate strength 0.25 (validation)	**40.062**	**4.134**

TABLE 11. Selected 168-Hour Paired Block-Bootstrap Ablations

Comparison	MSE diff.	Unadj. 95% CI low	Unadj. 95% CI high	Holm-adjusted p
M04 − M03: weather/calendar	-2.542	-6.695	0.940	1.000
M07 − M04: cosine retrieval features	-1.071	-2.280	-0.041	1.000
M08 − M07: event-conditioned ranking	0.151	-0.447	0.789	1.000
C04 − A02: event signal/placebo	-0.524	-0.983	-0.142	0.389
M09 − M08: drift features	0.107	-0.386	0.502	1.000
M13 − C06: TRACE residual gate	-0.052	-0.393	0.353	1.000
M13 − A03: TRACE event channel	-0.0001	-0.006	0.008	1.000
M13 − C05: proposed/best tree	-0.027	-0.350	0.353	1.000
M11 − M10: Chronos retrieval	1.075	-7.177	6.712	1.000
M12 − M04: drift router	-0.540	-1.112	-0.066	0.747

TABLE 12. Cosine Retrieval Sensitivity to *k*

k	MSE	MAE	RMSE
1	84.602	5.868	9.198
4	57.092	5.011	7.556
8	53.549	4.860	7.318
16	51.316	4.773	7.164

The density of the knowledge-base has a non-monotone effect. Table 13 indicates that feature-level event-aware and retrieval-only variants of the event-aware memory achieve 6-hour memory best, so admitting all hourly candidates is suboptimal. The results show that event-conditioned retrieval outperforms cosine retrieval by 1.669 MSE at stride 6 hours, but underperforms at 24- and 1-hour stride. This interaction encourages reliability gating, as opposed to an assumption that similarity of events is always good.

C. EVENT-SIMILARITY WEIGHT SENSITIVITY

The main event LSER coefficient ($\delta = 0.2$) was used in the configuration prior to final-test evaluation. Table 14 gives the full sweep performed with the time-series, weather and calendar coefficients fixed. The learned M08 validation MSE are only in the range 35.081 - 35.278 and less than 2.9% of test queries are affected by the change of event term in the set of retrieval candidates with respect to $\Delta = 0$. The learned M08 model is numerically lowest at $\delta = 1.0$ on validation and test while the lowest retrieval only event-context MSE is at the primary value. The fixed primary value is thus not alleged to be optimum. Most significantly, no stable event-specific accuracy gain is apparent in the sweep and the paper's qualified event conclusion remains the same.

D. EVENT AND DRIFT SUBSETS

Two pre-specified subsets which meet the 50-origin gate are reported in Table 15. Within the larger 72-hour event-context

TABLE 13. Knowledge-Base Density Sensitivity on the Final Holdout

Stride (h)	Candidates	Cosine MSE	Event-retrieval MSE	M08 MSE	M09 MSE
24	1,824	53.549	54.098	41.085	41.193
6	7,608	**49.524**	**47.855**	**40.520**	**40.886**
1	45,784	52.738	54.253	41.172	41.749

TABLE 14. Sensitivity to the LSER Event-Similarity Weight; the Primary Configuration is Marked and Test Values Were Not Used for Selection

Event weight δ	M08 validation MSE	M08 test MSE	Event-context retrieval MSE	Candidate change vs. $\delta = 0$ (%)
0.0	35.227	40.842	12.487	0.00
0.2 (primary)	**35.201**	**41.085**	**11.961**	**2.70**
0.5	35.278	40.690	12.151	2.78
0.8	35.110	40.858	13.735	2.79
1.0	35.081	40.620	14.322	2.83

set, there are 70 origins; direct target-event overlap contains 33 origins, so it is not scored. The mean and variance of event-context targets is 6.26 and 11.26, respectively, whereas for the rest of the targets it is 11.14 and 73.20. Thus, the magnitude of the MSE on a cross-subset is not regarded as the level of difficulty for this event. Even in the same origins of events, it is not possible to distinguish TRACE-RAF and its event-free ablation. The drift subset of 407 origins with the DFEH validation-calibrated threshold shows much higher target variance (105.34) and thus a high absolute error, partly due to that.

The composite detector detects 407 test origins at 0.2132 and a validation-calibrated two-sample KS benchmark detects 348 test origins at 0.4523. They have a Jaccard agreement of just 0.117. The composite flag, therefore, is regarded as a model-specific regime label, rather than a definition of drift, per se. The separately selected M12 router does likewise for M04, nominally – but not after Holm correction – which makes this cautious interpretation seem all the more appropriate.

E. SITE-LEVEL AND DECISION SENSITIVITY

The county median is the target of interest, since no monitor is at the 0.70 coverage gate. Three monitor-level arms were trained to evaluate if aggregation alone is the only determinant to event finding; no changes to the temporal protocol were made. Heterogeneous effects are seen in Table 16, with poorer performance at site 1103, and slightly better performance at sites 4009 and 9035. Event-only site subsets have between 10-33 origins and will not be scored. The checks are internal checks for target-sensitivity and not external geographic validation.

Only 37 of 2,460 test origins are exceedances at the 24-hour-mean threshold of 35.4 $\mu\text{g}/\text{m}^3$; none of the test origins exceed 55.4 $\mu\text{g}/\text{m}^3$. The recall of physical-threshold forecasts is very low. LightGBM achieves precision 0.234, recall 0.405, and F1 0.297 while TRACE-RAF achieves 0.213, 0.351, and 0.265, respectively, with validation calibrated decision thresholds. These sparse-event results are reported to link to the regression objective in air-quality decision-making, but these results are not adequate for a stand-alone alerting claim.

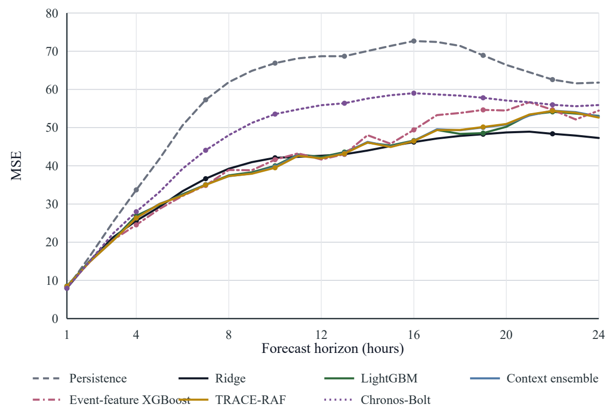

FIGURE 5. MSE by forecast horizon on the Los Angeles County test set.

F. HORIZON-WISE AND RETRIEVAL DIAGNOSTICS

Horizon-wise test error of comparators (principal supervised, feature retrieval and frozen TSFM and TRACE-RAF) is shown in Fig. 5 and Fig. 6. All models show a late-horizon failure, but no model has an isolated failure that would be disqualifying for the aggregate comparison since the ranking changes with lead time. The DFEH score and fixed threshold (DFT) are plotted on the test timeline in Fig. 7. Figures 8 and 9 show examples of ordinary and event-context figures, respectively, which are used to check rather than select models. All plots are regenerated from run 20260827T043457543402Z.

G. FORECAST EVIDENCE AND AUDITABILITY ANALYSIS

In the DFEH, a source-grounded forecast evidence record is kept and not a full causal explanation. The pipeline stores the run and window identifiers, the origin time, retrieved historical windows, linked event records, leading feature effects and drift components, retrieval spread, validation residual RMSE, uncertainty proxy, drift flag, and event-availability mode for each of the 2,460 test origins. The

TABLE 15. Representative Event and Drift Subset Results

Subset	Model	Origins	MSE	MAE
Event context	C01 Ridge context	70	5.624	1.925
Event context	C06 context ensemble	70	6.568	2.101
Event context	M13 TRACE-RAF	70	6.552	2.097
Event context	A03 TRACE without events	70	6.553	2.095
Non-drift	C01 Ridge context	2,053	38.808	3.989
Non-drift	M13 TRACE-RAF	2,053	38.085	3.955
Drift	C01 Ridge context	407	40.843	4.536
Drift	C06 context ensemble	407	50.267	5.077
Drift	M13 TRACE-RAF	407	50.033	5.039
Drift	A03 TRACE without events	407	50.047	5.039

TABLE 16. Monitor-Level Event-Retrieval Sensitivity

Site	Coverage	Origins	M04 MSE	M08 MSE
06-037-1103	0.558	1,631	68.281	69.898
06-037-4009	0.510	1,732	73.506	73.095
06-037-9035	0.374	547	17.682	17.371

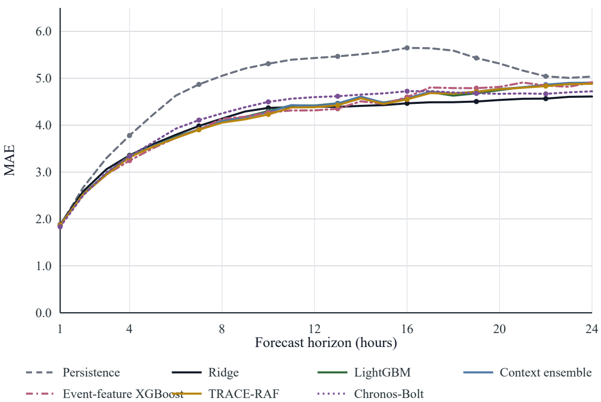

FIGURE 6. MAE by forecast horizon on the Los Angeles County test set.

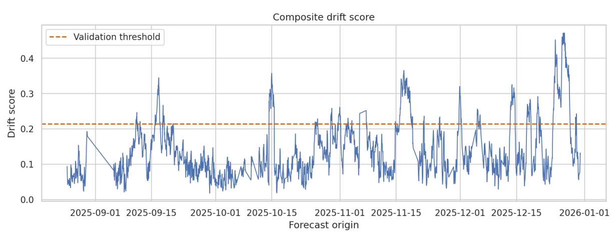

FIGURE 7. Composite drift score and the validation-calibrated threshold on the final holdout. The flag defines a diagnostic regime rather than verified distribution-shift ground truth.

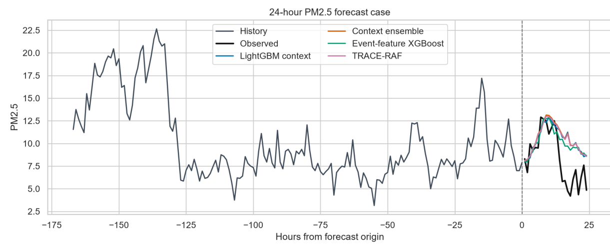

FIGURE 8. Representative 24-hour forecast from the final holdout. Model traces are plotted against the observed hourly target without post-test adjustment.

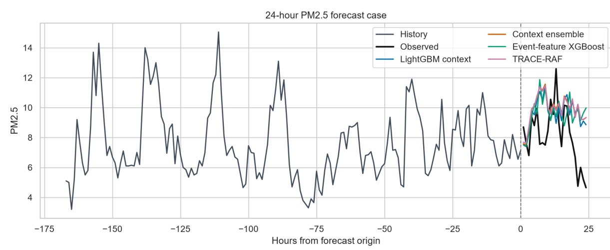

FIGURE 9. Representative forecast whose target lies within the declared 72-hour event-context window. The event designation is retrospective because Storm Events files do not expose publication timestamps.

TABLE 17. Lossless TRACE Forecast-Contribution Decomposition

Subset	Origins	Mean g	Mean $\|\Delta\|$	Mean $\|g\Delta\|$
All	2,460	0.090	1.708	0.149
Event context	70	0.082	0.736	0.060
Non-event context	2,390	0.090	1.736	0.152
Distribution shift	407	0.076	2.386	0.175
No distribution shift	2,053	0.093	1.573	0.144

`explanations.parquet` file thus corresponds 1:1 to the forecasts that have been evaluated. It's main local fields are typically current PM2.5, and recent rolling concentrations, and analogue and event identifiers reveal information for the correction mechanism.

The evidence layer is summarized in Table 18. This design keeps track of the auditability of each evidence statement by linking the evidence statement to either a model feature, a retrieved time-series window, an official event record, or a drift component. It's not a proof of causality: event evidence is presented as contextual support and drift flag is used as a warning of uncertainty. Faithfulness is assessed by perturbing entire families of features in the model used for the fit (M09 model). As seen in Table 19, the history of PM2.5 is by far the most important one, followed by weather and calendar information. Permutations of events and retrievals lead to an increase in MSE, and a negative median-deletion effect. These mixed signs preclude a more ambitious assertion: each

TABLE 18. Forecast Evidence Stored by TRACE-RAF

Evidence field	Interpretation role
Top feature effects	Local model drivers from the direct forecaster.
Retrieved windows	Historical analogues used by the retrieval module.
Event evidence IDs	Official storm-event context linked to the origin.
Drift components	Mean, variance, weather, similarity, and event-burst shift signals.
Uncertainty proxy	Combined retrieval spread and validation residual error.

TABLE 19. M09 Feature-Family Faithfulness Perturbations

Feature family	Permutation ΔMSE	Median deletion ΔMSE
PM2.5 history	25.575	13.531
Weather	3.440	3.089
Calendar	3.548	2.481
Events	1.162	-0.442
Retrieval	0.964	-0.551
Drift	0.268	0.125

type of stored evidence is a stable causal explanation.

The lossless prediction archive also allows to reconstruct the final TRACE update $\hat{Y}^{(\mathrm{TRACE})} = \hat{Y}^{B} + g\Delta$ directly. The effective origin-level coefficient g is applied to this value as a result of the validation selection of strength of 0.25. The quantity of each component is quantified in Table 17. The learned gate leads to a mean absolute raw residual analogue of 1.708, whereas the mean absolute applied correction is only 0.149 $\mu\mathrm{g}/\mathrm{m}^3$ for all the test origins. The mean applied correction for event-context origins is the smallest, 0.060 $\mu\mathrm{g}/\mathrm{m}^3$. These values reveal the information that went into determining the final forecast value; they are not causal attribution to the event records.

H. COMPUTATIONAL COST AND REPRODUCIBILITY

The publication pipeline had an internal runtime of 7432.7 seconds in a Linux environment with Python 3.12.13. All the three notebooks done correctly without cell errors. The recorded versions include NumPy 2.0.2, pandas 2.3.3, scikit-learn 1.6.1, XGBoost 3.2.0, LightGBM 4.6.0, PyTorch 2.10.0+cu128, Transformers 4.57.6, and chronos-forecasting 2.3.1. There are 43 result tables in the archive, along with one run identifier, configuration and artifact hashes, prediction level outputs, evidence records and an independent claim verification notebook. The stored subset inference predictions are matched up to a max abs error of 9.85×10^{-17} with the TRACE subset-inference reconstruction. No hardware identifiers or Git commit were found in the notebook environment, so the runtime is not considered to be a hardware-normalized efficiency result.

Our primary interpretation of this is that the supervised context models are quite strong for this Los Angeles County task. Although context XGBoost and TRACE-RAF have lower errors, no favourable TRACE contrasts survive family-wise adjustment and Ridge is best overall in both MSE and MAE. The mechanism ladder additionally demonstrates that validation rejects raw and globally scaled analogue fusion, while the learned gate applies only a small fraction of the retrieved residual. The result of the whole TRACE, which is almost identical to the event-free ablation, is due to conservative residual correction and not to measurable conditioning by events. In contrast, the addition of event retrieval to frozen Chronos-Bolt causes a huge degradation in MAE after Holm correction. The resulting boundary is thus a reproducible negative and positive one: trust-gated residual analogues are competitive, as is event-conditioned output-space fusion with frozen Chronos-Bolt-small, yet a holdout does not support this latter boundary. This score does not reflect the general performance of internal TSFM prompting nor retrieval-augmented TSFMs.

TABLE 20. Computational and Reproducibility Summary

Item	Recorded value
Run identifier	`20260827T043457543402Z`
Pipeline runtime	7,432.7 seconds
Platform	Linux 6.12.90
Python	3.12.13
Forecast horizon	24 hours
Lookback window	168 hours
Bootstrap resamples	5,000 in 168-hour blocks
Multiple testing	Holm adjustment over the comparison family
Chronos checkpoint	`amazon/chronos-bolt-small`
Code availability	https://github.com/shasabbir/event-time-raf

There is a pre-specified stress analysis for January 2025, with 622 origins, and is only included as development evidence as it is part of the validation period. Ridge, Chronos-Bolt, the context ensemble, and TRACE-RAF obtain MSE 118.300, 116.012, 123.305, and 123.643, respectively. It is not an extra test set, and does not affect the final ranking of the model.

Lastly and most importantly, it is necessary to read all event-aware interpretations with an availability disclaimer. NOAA Storm Events are hash verified and official, but the timeline of publication is not provided in machine-readable format. The availability time is set as the event start in the experiment and all results relating to the event are retrospective sensitivity results. In order to perform a strict operational study, it would be necessary to use an event or smoke source with "true" issue or publication times, such as an audited NOAA HMS cache or alert product archive.

IX. LIMITATIONS

The first restriction is on the availability of events. The NOAA Storm Events detail files are officially verified and using hash but the publication time is not machine-readable from the detail files. The SACB therefore considers the results in event-conditioned scenarios as retrospective sensitivity analysis and not operational forecasts; the availability proxy is therefore based on the event start.

Geographic validation is the second limitation. There is one county-level PM2.5 aggregate for the experiment 2019-2025. The monitor level analysis suggests that there is a change of sign across sites but not necessarily a transfer to another county, climate or emissions regime.

The third restriction is the make-up of the holdout. Only 33 target-event origins and 37 threshold exceedances are observed and no test origin exceeds 55.4 $\mu g/m^3$. The qualifying category analysis spans the larger 70-origin event-context subset, which includes flood, heavy-rain, and other-weather contexts, but not a wildfire, smoke, or high-wind query. The high-error period in January 2025 is not an external confirmation, but validation evidence. Reports of serious smoke episodes or alerts to the operation would thus be more than what is supported by observations of the tests.

The fourth restriction is on targets building. A county median minimizes the number of missing monitors, flattens out local peaks and is dependent on the number of monitors that are active. The lower-coverage monitor arms show some level of heterogeneity but are not dense enough to perform event-subset inference.

The fifth restriction is with regard to the empirical power. The performance of TRACE-RAF is not better than Ridge and both the residual-gate contrast and the event-channel contrast are not significant after Holm adjustment. There is no faithfulness perturbation on evidence records, they are auditable, but the faithfulness perturbations for the event and retrieval families are mixed and are not useful for causal interpretation.

The 6th limitation is model integration. LSER is an open similarity retriever, not a learned dual encoder and Chronos-Bolt is an external benchmark, not an internal backbone for prompting. It should not be marketed as an alternate of the TimeRAF Channel Prompting mechanism.

Lastly, there were no hardware identifiers, peak memory or a Git commit inside the extracted run environment. Package versions, configuration hashes, archive hashes and prediction-level verification allow for more reproducible runs, but the reported runtimes are not hardware-normalized efficiency figures.

X. FUTURE WORK

Operational event studies should use archives that provide real time of publication/issue of the event and, if the event transport mechanism is a wildfire, should add smoke-specific sources to replace the event-start proxies. External validation should be: application of the frozen protocol to a second US county with different meteorology and emissions and monitor density. Then, spatial or site-aware forecasting can be used to investigate whether event effects are hidden in the aggregation over counties. Future research could assess a learned residual retriever, more expressive event embedding, an internally conditioned TSFM and calibrated exceedance probabilities. The extension needs to be pre-registered with a third party holdout, and components need to be ablated under the same time embargo.

XI. CONCLUSION

In this study, TRACE-RAF, a Trust-Gated Residual Analog Correction approach for auditable 24-hour PM2.5 forecasting is introduced. SACB keeps the official records of PM2.5, GHCNh weather, calendar, and Storm Events; LSER enforces a strict historical embargo; TRACE retrieves out-of-fold residual analogues and applies a learned reliability gate; and DFEH provides prediction-level evidence and drift diagnostics. On the final 2025 holdout of 2,460 origins, TRACE-RAF obtains MSE 40.062, MAE 4.134, RMSE 6.329, and $R^2 = 0.444$. This marks the lowest error of the context XGBoost and nominally beats the LightGBM comparator and Ridge is still best at MSE 39.144. Both raw-future fusion and a global residual scale are rejected by the mechanism ladder, the learned gate provides a conservative reduction in the residual update and gives a nominal MAE improvement. No favorable TRACE comparison survives the Holm adjustment and the event-free TRACE ablation is essentially the same. The supported contribution is thus a reproducible, trust-gated residual-retrieval architecture and an auditable assessment of event-conditioned residual retrieval and a single output-fusion (frozen Chronos-Bolt-small) configuration, not about retrieval-augmented TSFMs in general or universal baseline superiority. Event issue times, geographic testing outside of the system and smoke specific validation are still required prior to operational deployment.

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